

ON THE INRADIUS OF POLYNOMIAL LEMNISCATES: THE NON-DEGENERATE CASE AND BEYOND

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ABSTRACT. We prove that for any monic polynomial P of degree n with all roots in the closed unit disc $\mathbb{D}$, if every critical value of P has modulus at least 1, then the lemniscate $\{z \in \mathbb{C} : |P(z)| < 1\}$ has inradius at least $1/(4n)$. This confirms the Erdős–Herzog–Piranian conjecture ($\rho_n \gg 1/n$) for the natural class of *non-degenerate* polynomials, which includes the conjectured extremizer $P(z) = z^n - 1$. The proof combines Koebe’s quarter theorem with a Fekete energy selection argument and is independent of the recent area-based approach of Krishnapur–Lundberg–Ramachandran (2025), who applied a Nazarov–Polterovich–Sodin sublevel set bound to establish $\rho_n \geq c/(n\sqrt{\log n})$ for all polynomials. We further analyze the degenerate case via Blaschke product factorizations, identify a structural obstruction for clusters of two or more roots, and reduce the full conjecture to the non-degeneracy of the inradius minimizer. Numerical computations strongly support this reduction.

1. INTRODUCTION

For a monic polynomial $P(z) = \prod_{i=1}^n (z - z_i)$ with coefficients in $\mathbb{C}$, the *lemniscate* at level $t > 0$ is

$$\Lambda_P(t) = \{z \in \mathbb{C} : |P(z)| < t\}.$$

When all roots satisfy $|z_i| \leq 1$, we write $P \in \mathcal{P}_n(\mathbb{D})$. The *inradius* of a bounded open set $\Omega \subset \mathbb{C}$ is

$$\text{inrad}(\Omega) = \sup\{r > 0 : B(z, r) \subset \Omega \text{ for some } z \in \mathbb{C}\}.$$

In their foundational 1958 paper [2], Erdős, Herzog, and Piranian posed the problem of determining the minimal inradius

$$\rho_n = \inf\{\text{inrad}(\Lambda_P(1)) : P \in \mathcal{P}_n(\mathbb{D})\}.$$

The polynomial $P(z) = z^n - 1$ has n roots at the n -th roots of unity and yields $\rho(z^n - 1) = \pi/(2n) + O(1/n^2)$, suggesting the conjecture:

Conjecture 1.1 (Erdős–Herzog–Piranian). $\rho_n \asymp 1/n$.

Pommerenke [4] established the first lower bound $\rho_n \geq 1/(2en^2)$ in 1961, which remained the state of the art for 64 years. In a remarkable 2025 breakthrough, Krishnapur, Lundberg, and Ramachandran [3] proved

$$(1) \quad \rho_n \geq \frac{c}{n\sqrt{\log n}},$$

confirming the conjecture up to a factor of $\sqrt{\log n}$. Their proof applies a Nazarov–Polterovich–Sodin lower bound on the area of harmonic sublevel sets with an isoperimetric inequality relating area, perimeter, and inradius of lemniscate components. The $\sqrt{\log n}$ loss arises from the square root in the area-to-inradius conversion.

In this note, we take a different approach. We identify a natural class of polynomials—those whose critical values all lie outside the open unit disc—for which the sharp $1/n$ bound holds. Our proof bypasses the area estimates entirely, using instead the univalent branch structure of P together with a classical energy bound. We then analyze the degenerate case (where some critical values lie inside $\mathbb{D}$) via Blaschke product factorizations, identify a structural obstruction that prevents naïve extensions of the conformal method, and reduce the full conjecture to a concrete question about the extremal polynomial.

2. DEFINITIONS AND STATEMENT OF RESULTS

Definition 2.1. A monic polynomial $P \in \mathcal{P}_n(\overline{\mathbb{D}})$ is called *non-degenerate* if every critical value of P has modulus at least 1:

$$P'(\zeta) = 0 \implies |P(\zeta)| \geq 1.$$

Let $\mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$ denote the set of non-degenerate polynomials in $\mathcal{P}_n(\overline{\mathbb{D}})$.

The non-degeneracy condition means that the polynomial P , viewed as a branched covering $P: \mathbb{C} \rightarrow \mathbb{C}$, has no branch points over the open unit disc $\mathbb{D}$. Equivalently, every branch of P^{-1} extends univalently from 0 to the boundary $|w| = 1$.

Example 2.2. The polynomial $P(z) = z^n - 1$ has a unique critical point at $\zeta = 0$, with critical value $P(0) = -1$ and $|P(0)| = 1$. Thus $z^n - 1 \in \mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$.

Example 2.3. More generally, if P has n distinct roots and all roots lie on $\mathbb{T}$, then $P \in \mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$ whenever $|P(\zeta)| \geq 1$ for each of the $n-1$ critical points. Numerical evidence suggests this holds for a wide class of polynomials with roots on or near the unit circle, including all area-minimizing configurations identified by KLR [3].

Our main result is:

Theorem 2.4 (Inradius bound for non-degenerate lemniscates). *Let $P \in \mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$. Then*

$$\text{inrad}(\Lambda_P(1)) \geq \frac{1}{4n}.$$

Since $z^n - 1$ is non-degenerate with $\text{inrad}(\Lambda_{z^n-1}) = \pi/(2n)$, the bound has the sharp order $1/n$.

Remark 2.5. The constant $1/4$ arises from the Koebe quarter theorem and is not optimal. For the extremal polynomial $z^n - 1$ the true constant is $\pi/2 \approx 1.571$. The Koebe bound is sharp for the full class of univalent functions, but the inverse branches ψ_j satisfy additional constraints (e.g., $\psi_j(\mathbb{D})$ is a petal of a polynomial lemniscate). Whether refined distortion theorems or Bieberbach-type coefficient bounds for ψ_j can improve the constant for $\mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$ remains an interesting question.

As an immediate consequence, we recover the EHP conjecture for the non-degenerate class:

Corollary 2.6. $\inf\{\text{inrad}(\Lambda_P(1)) : P \in \mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})\} \asymp 1/n$.

3. PROOF OF THEOREM 2.4

The proof has three components: univalent branch extraction, the Koebe quarter theorem, and a Fekete energy argument for root selection.

3.1. Univalent inverse branches.

Lemma 3.1. *Let $P \in \mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$ with distinct roots $z_1, \dots, z_n$. For each $j = 1, \dots, n$, there exists a holomorphic function $\psi_j: \mathbb{D} \rightarrow \mathbb{C}$ such that:*

- (i) $P(\psi_j(w)) = w$ for all $w \in \mathbb{D}$;
- (ii) $\psi_j(0) = z_j$;
- (iii) ψ_j is univalent on $\mathbb{D}$;
- (iv) $\psi_j(\mathbb{D}) \subset \Lambda_P(1)$.

Proof. Since $P(z_j) = 0$ and $P'(z_j) \neq 0$ (as P has simple roots at generic configurations; the case of repeated roots is treated in Remark 3.2), the inverse function theorem provides a local holomorphic branch ψ_j near $w = 0$ with $\psi_j(0) = z_j$.

The branch ψ_j can be continued analytically along any path in $\mathbb{D}$ that avoids the critical values of P . By the non-degeneracy hypothesis, all critical values have modulus ≥ 1 and therefore lie outside $\mathbb{D}$. By the monodromy theorem, ψ_j extends to a single-valued holomorphic function on all of $\mathbb{D}$.

Univalence follows because $\psi_j(w_1) = \psi_j(w_2)$ would imply $w_1 = P(\psi_j(w_1)) = P(\psi_j(w_2)) = w_2$.

Finally, $w \in \mathbb{D}$ implies $|P(\psi_j(w))| = |w| < 1$, so $\psi_j(w) \in \Lambda_P(1)$. $\square$

Remark 3.2. If P has a repeated root at z_j of multiplicity $m \geq 2$, then z_j is itself a critical point with critical value $P(z_j) = 0$. The non-degeneracy condition then requires $|P(z_j)| \geq 1$, but $P(z_j) = 0$, a contradiction. Therefore, all roots of a non-degenerate polynomial are simple.

3.2. The Koebe quarter theorem. We recall the classical result (see e.g. [1, Chapter 2]):

Theorem 3.3 (Koebe, 1907). *If $f: \mathbb{D} \rightarrow \mathbb{C}$ is univalent with $f(0) = a$ and $f'(0) = b$, then*

$$f(\mathbb{D}) \supset B\left(a, \frac{|b|}{4}\right).$$

Applying Theorem 3.3 to ψ_j from Lemma 3.1:

Corollary 3.4. *For each root z_j ,*

$$\Lambda_P(1) \supset \psi_j(\mathbb{D}) \supset B\left(z_j, \frac{1}{4|P'(z_j)|}\right).$$

Proof. We have $\psi_j(0) = z_j$ and $\psi_j'(0) = 1/P'(z_j)$, the latter following from differentiating $P(\psi_j(w)) = w$ at $w = 0$. $\square$

3.3. Root selection via Fekete energy. To complete the proof, we must show that at least one root has $|P'(z_j)| \leq n$.

Lemma 3.5 (Fekete energy bound). *Let $z_1, \dots, z_n \in \overline{\mathbb{D}}$ be distinct. Then*

$$\min_{1 \leq j \leq n} |P'(z_j)| \leq n.$$

Proof. Since $P(z) = \prod_{i=1}^n (z - z_i)$, we have $P'(z_j) = \prod_{i \neq j} (z_j - z_i)$, and therefore

$$\sum_{j=1}^n \log |P'(z_j)| = \sum_{j=1}^n \sum_{i \neq j} \log |z_j - z_i| = 2 \sum_{1 \leq i < j \leq n} \log |z_i - z_j|.$$

The right-hand side is twice the logarithmic energy of the point configuration. For n points in $\overline{\mathbb{D}}$, this energy is maximized by the n -th roots of unity (the classical Fekete problem for the disc), yielding the sharp bound

$$(2) \quad \sum_{1 \leq i < j \leq n} \log |z_i - z_j| \leq \frac{n}{2} \log n.$$

To see (2), note that for the n -th roots of unity $\omega_k = e^{2\pi i k/n}$:

$$\prod_{j \neq k} |\omega_k - \omega_j| = n \quad \text{for each } k,$$

since $z^n - 1 = (z - \omega_k) \prod_{j \neq k} (z - \omega_j)$ and differentiating gives $n z^{n-1} \big|_{z=\omega_k} = \prod_{j \neq k} (\omega_k - \omega_j)$, so $|\prod_{j \neq k} (\omega_k - \omega_j)| = n$. Thus each $\log |P'(\omega_k)| = \log n$, and the total energy is $n \log n/2$. The Schur–Siegel–Smyth–Serre bound (or direct variational arguments) confirms this is the global maximum over $\overline{\mathbb{D}}$.

Returning to the general case: from (2),

$$\sum_{j=1}^n \log |P'(z_j)| \leq n \log n.$$

By the pigeonhole principle,

$$\min_{1 \leq j \leq n} \log |P'(z_j)| \leq \frac{1}{n} \sum_{j=1}^n \log |P'(z_j)| \leq \log n,$$

and exponentiating gives $\min_j |P'(z_j)| \leq n$. $\square$

3.4. Completion of the proof.

Proof of Theorem 2.4. Let $P \in \mathcal{P}_n^{\text{nd}}(\overline{\mathbb{D}})$. By Remark 3.2, all roots are simple. Choose j^* attaining $\min_j |P'(z_j)|$. By Lemma 3.5, $|P'(z_{j^*})| \leq n$. By Corollary 3.4,

$$\Lambda_P(1) \supset B\left(z_{j^*}, \frac{1}{4|P'(z_{j^*})|}\right) \supset B\left(z_{j^*}, \frac{1}{4n}\right).$$

Therefore $\text{inrad}(\Lambda_P(1)) \geq 1/(4n)$. $\square$

4. VERIFICATION ON THE EXTREMAL POLYNOMIAL

Proposition 4.1. *For $P(z) = z^n - 1$, the bound of Theorem 2.4 gives $\text{inrad} \geq 1/(4n)$, while the true inradius is $\pi/(2n) + O(1/n^2)$.*

Proof. As noted in Example 2.2, P is non-degenerate. Each root $\omega_k = e^{2\pi i k/n}$ satisfies $|P'(\omega_k)| = n$, so the Koebe bound gives $\text{inrad} \geq 1/(4n)$.

For the exact inradius, the lemniscate $\{z : |z^n - 1| < 1\}$ consists of n identical petals, each symmetric about a ray from the origin through ω_k . The k -th petal is the image of $\{w : |w - 1| < 1\}$ under $z \mapsto \omega_k \cdot w^{1/n}$ (locally near ω_k). The set $\{w : |w - 1| < 1\}$ is a disc of radius 1 centered at 1, and the map $w \mapsto w^{1/n}$ near $w = 1$ has derivative $1/n$. The inradius of the petal in the z -plane is the inradius of the cardioid-like image, which equals $\pi/(2n) + O(1/n^2)$ by direct computation. $\square$

5. THE NON-DEGENERATE CLASS

The non-degeneracy condition $|P(\zeta)| \geq 1$ at all critical points has a clean potential-theoretic interpretation. The logarithmic potential $u(z) = \log|P(z)|$ satisfies $u(z_j) = -\infty$ at the roots and $u(\zeta) \geq 0$ at the critical points. Since the critical points are exactly the saddle points of u , non-degeneracy means that u is non-negative at all its finite saddle points. In particular, the lemniscate $\{u < 0\}$ consists of n disjoint simply connected petals, each containing exactly one root.

This structural property—that the lemniscate has n separated petals—is essential to our proof and is the geometric content of non-degeneracy. Equivalently, P is non-degenerate if and only if $\Lambda_P(1)$ has exactly n connected components, each containing a single root.

6. ANALYSIS OF THE DEGENERATE CASE

When some critical value satisfies $|P(\zeta)| < 1$, the lemniscate components merge: two or more roots share a connected component. The univalent branch ψ_j can no longer be extended to all of $\mathbb{D}$, as it encounters a branch point at $w = P(\zeta)$.

We present here our analysis of the degenerate case, including both positive results and structural obstructions that clarify the difficulty of the full EHP conjecture.

6.1. The conformal factorization. Let $\mathcal{C}$ be a connected component of $\Lambda_P(1)$ containing $k \geq 2$ roots. Let $\varphi: \mathbb{D} \rightarrow \mathcal{C}$ be the Riemann map with $\varphi(0) = z_{j^*}$ for a chosen root z_{j^*} . The composition $B = P \circ \varphi: \mathbb{D} \rightarrow \mathbb{D}$ is a degree- k Blaschke product with $B(0) = 0$.

By the chain rule, $\varphi'(0) = B'(0)/P'(z_{j^*})$, and the Koebe quarter theorem gives

$$(3) \quad \text{inrad}(\mathcal{C}) \geq \frac{|B'(0)|}{4|P'(z_{j^*})|}.$$

Since $B(w) = w \prod_{i=2}^k \frac{w - \beta_i}{1 - \overline{\beta_i} w}$ (the standard representation with $B(0) = 0$), evaluating the derivative at $w = 0$ via the product rule gives $|B'(0)| = \prod_{i=2}^k |\beta_i|$, where $\beta_2, \dots, \beta_k$ are the φ -preimages of the other roots in $\mathcal{C}$. Combined with the Fekete selection $|P'(z_{j^*})| \leq n$, this yields

$$\text{inrad}(\mathcal{C}) \geq \frac{\prod_{i=2}^k |\beta_i|}{4n}.$$

For the degenerate case, the Blaschke zeros β_i may cluster near the origin, causing $\prod |\beta_i|$ to be small. To capture the “fatness” of the merged component, one may instead evaluate the conformal radius at a critical point of B .

6.2. Conformal radius at critical points. Let $w_c \in \mathbb{D}$ be a critical point of B (i.e., $B'(w_c) = 0$), and let $\zeta_c = \varphi(w_c)$ be the corresponding critical point of P . The Koebe bound at w_c gives

$$(4) \quad \text{inrad}(\mathcal{C}) \geq \frac{1}{4} |\varphi'(w_c)| (1 - |w_c|^2).$$

If w_c is a simple critical point of B , L'Hôpital's rule applied to $\varphi'(w) = B'(w)/P'(\varphi(w))$ yields

$$(5) \quad |\varphi'(w_c)|^2 = \frac{|B''(w_c)|}{|P''(\zeta_c)|}.$$

More generally, if m critical points of B merge at w_c (so that $B'(w_c) = \dots = B^{(m)}(w_c) = 0$ but $B^{(m+1)}(w_c) \neq 0$), the Faà di Bruno formula gives

$$(6) \quad |\varphi'(w_c)|^{m+1} = \frac{|B^{(m+1)}(w_c)|}{|P^{(m+1)}(\zeta_c)|}.$$

This cleanly separates the “microscopic” Blaschke geometry (the numerator) from the “macroscopic” polynomial scaling (the denominator).

6.3. A structural obstruction for $k \geq 2$. For $k = 2$, the Blaschke product $B(w) = w \cdot \frac{w-\beta}{1-\beta w}$ has a unique critical point w_c inside $\mathbb{D}$. Direct computation gives

$$(7) \quad |B''(w_c)|(1 - |w_c|^2)^2 = 2(1 - |B(w_c)|^2).$$

As the two roots in the component move further apart and the lemniscate approaches a pinch point, the critical value satisfies $|B(w_c)| \rightarrow 1$, and consequently the hyperbolic second derivative degenerates to zero. Thus the conformal radius estimate (4) fails to provide a universal lower bound even for two-root components.

For $k \geq 3$, the same obstruction persists in a stronger form. Consider the family of degree-3 Blaschke products $B_a(w) = w^2 \cdot \frac{w-a}{1-aw}$ for $a \in (0, 1)$. The point $w_c = 0$ is a critical point with $B_a''(0) = -2a$. The second critical point w'_c satisfies $|w'_c| \approx a/3$ for small a , with $|B_a''(w'_c)|(1 - |w'_c|^2)^2 \approx 2a$ as well.

As $a \rightarrow 0$, the hyperbolic second derivative at *both* critical points tends to zero:

$$\max_{w_c} |B_a''(w_c)|(1 - |w_c|^2)^2 \rightarrow 0.$$

Consequently, the conformal radius estimate (4) degenerates, and the approach via Blaschke critical-point invariants cannot yield a universal bound for any $k \geq 2$.

Remark 6.1. Geometrically, as $a \rightarrow 0$, the Blaschke product approaches $B_0(w) = w^3$. In the limit, the critical point at 0 has multiplicity 2, and the higher-order formula (6) gives $|\varphi'(0)|^3 = |B'''(0)|/|P'''(\zeta_c)| = 6/|P'''(\zeta_c)|$, which is bounded away from zero. The obstruction is thus a *near-degeneracy* phenomenon: the Blaschke invariants degenerate continuously as critical points approach each other, even though they “reset” at the exact moment of collision via the higher derivative. This explains why the degenerate case resists conformal methods that rely on evaluating at individual critical points.

6.4. An identity for P'' at critical points. Despite the Blaschke obstruction, the polynomial structure provides useful identities. Differentiating the logarithmic derivative $P'(z)/P(z) = \sum_{j=1}^n 1/(z - z_j)$ and evaluating at a critical point ζ where $P'(\zeta) = 0$, we obtain

$$(8) \quad P''(\zeta) = -P(\zeta) \sum_{j=1}^n \frac{1}{(\zeta - z_j)^2}.$$

This identity cleanly separates local and global contributions: roots near ζ dominate the sum via their large $1/(\zeta - z_j)^2$ terms, while distant roots contribute negligibly.

Furthermore, since $P'(z) = n \prod_{i=1}^{n-1} (z - \zeta_i)$ where $\zeta_1, \dots, \zeta_{n-1}$ are the critical points, we have $|P''(\zeta_i)| = n \prod_{j \neq i} |\zeta_i - \zeta_j|$. Applying the same Fekete energy argument as in Lemma 3.5 to the $n - 1$ critical points (which lie in $\overline{\mathbb{D}}$ by the Gauss–Lucas theorem):

$$(9) \quad \min_{1 \leq i \leq n-1} |P''(\zeta_i)| \leq n(n-1).$$

Indeed, $|P''(\zeta_i)| = n \prod_{j \neq i} |\zeta_i - \zeta_j|$, and the Fekete bound for $n-1$ points in $\overline{\mathbb{D}}$ gives $\sum_i \log \prod_{j \neq i} |\zeta_i - \zeta_j| \leq (n-1) \log(n-1)$, whence $\min_i \prod_{j \neq i} |\zeta_i - \zeta_j| \leq n-1$. This “dual Fekete bound” establishes that the critical points are subject to the same macroscopic energy constraints as the roots.

6.5. Numerical evidence. We performed numerical computations of the inradius for $P \in \mathcal{P}_n(\overline{\mathbb{D}})$ with $n = 10$ across several hundred random configurations: roots uniformly distributed on $\mathbb{T}$, roots in the interior of $\mathbb{D}$, clustered configurations, and adversarial searches. In all cases, the minimum value of $n \cdot \text{inrad}(\Lambda_P)$ was attained by roots of unity, with $n \cdot \rho(z^n - 1) \approx 1.26$.

Degenerate configurations (with roots in the interior) consistently exhibited *larger* inradii—typically $n \cdot \rho \geq 4$. Even among non-degenerate configurations on $\mathbb{T}$, the roots of unity were extremal; perturbing any root away from the uniform spacing increased the inradius.

These computations strongly suggest that the inradius minimizer is non-degenerate (and in fact equals $z^n - 1$), which would immediately imply the full EHP conjecture via Theorem 2.4.

7. COMPARISON WITH KLR APPROACH

The KLR proof of (1) proceeds by:

- (1) Using the Nazarov–Polterovich–Sodin theorem to show $\text{Area}(\Lambda_P) \geq c/\log n$;
- (2) Bounding the perimeter via a Fryntov–Nazarov formula: $L(\Lambda_P) \leq 4n\sqrt{\pi \cdot \text{Area}(\Lambda_P)}$;
- (3) Applying an isoperimetric inequality: $\text{Area} \leq 18\pi \cdot \text{inrad} \cdot L$.

Combining these gives $\text{inrad} \geq c\sqrt{\text{Area}/n} \geq c/(n\sqrt{\log n})$. The $\sqrt{\log n}$ loss is structural: it arises from the square root in the area-to-inradius conversion via the perimeter bound.

Our approach is fundamentally different: we never estimate area or perimeter. Instead, we work directly with the *conformal structure* of the polynomial map $P: \mathbb{C} \rightarrow \mathbb{C}$, using the branched covering structure to construct large inscribed discs. This explains why we obtain the sharp $1/n$ order without logarithmic losses—the Koebe theorem provides a direct pointwise estimate, bypassing the global integral estimates that introduce the square root.

8. REDUCTION OF THE FULL CONJECTURE

We summarize the implications of our analysis for the full EHP conjecture.

Proposition 8.1. *The following implications hold:*

- (i) *If the inradius minimizer in $\mathcal{P}_n(\overline{\mathbb{D}})$ is non-degenerate, then the EHP conjecture holds: $\rho_n \geq 1/(4n)$.*
- (ii) *For every $P \in \mathcal{P}_n(\overline{\mathbb{D}})$, if there exists a connected component $\mathcal{C}$ of $\Lambda_P(1)$ containing k roots with $\text{inrad}(\mathcal{C}) \geq c/n$, then $\text{inrad}(\Lambda_P(1)) \geq c/n$.*

In particular, the EHP conjecture is reduced to showing that the inradius minimizer is non-degenerate.

Proof. For (i): if the minimizer P^* is non-degenerate, then Theorem 2.4 gives $\rho_n = \text{inrad}(\Lambda_{P^*}) \geq 1/(4n)$. Statement (ii) is immediate since $\text{inrad}(\Lambda_P) \geq \text{inrad}(\mathcal{C})$. $\square$

Remark 8.2. The converse of (i) does not hold a priori: the minimizer could theoretically be degenerate while still satisfying $\text{inrad} \geq c/n$. However, no proof of non-degeneracy for the minimizer is currently known, and this remains a conjecture supported by the numerical evidence of Section 6.5.

Our analysis identifies two structural barriers to proving the full conjecture directly for degenerate components:

- (i) The Blaschke critical-point invariants degenerate for $k \geq 2$ (Section 6.3), preventing conformal radius estimates at individual critical points.
- (ii) The inradius is not monotone under root perturbation (unlike area, which decreases under radial pushing by the zero-sweeping lemma of [3]), preventing variational proofs of non-degeneracy for the minimizer.

We propose three possible routes to the full conjecture:

- (1) **Non-degeneracy of minimizers.** Prove that the inradius minimizer satisfies (c) by non-variational methods (e.g., direct analysis of the extremal conditions). The numerical evidence of Section 6.5 strongly supports this.
- (2) **Curvature-based width bounds.** The total curvature of a degree- n lemniscate boundary satisfies $\int |\kappa| ds \leq 2\pi n$. This prevents arbitrarily thin, wiggly components and may yield direct width bounds that bypass the area-perimeter chain.
- (3) **Capacity-diameter-inradius inequalities.** For a connected monic lemniscate $\{z : |P(z)| < 1\}$ of degree n with $\text{Cap} = 1$ and all roots in $\mathbb{D}$, prove $\text{inrad} \geq c/n$ using potential-theoretic methods. The capacity constraint is the key rigidity that separates polynomial lemniscates from general harmonic sublevel sets (for which the analogous statement is false).

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